Understanding the Impact of Principal Component Selection on Data Representation and Model Performance

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Introduction

In modern machine learning, datasets often contain a large number of features, making them complex, computationally expensive, and difficult to interpret. As the number of dimensions increases, models may struggle to identify meaningful patterns, leading to reduced performance. **Dimensionality reduction** is a technique used to simplify such datasets by reducing the number of input variables while preserving the most important information. This makes data easier to visualise, speeds up computation, and can improve model accuracy.

One of the most widely used dimensionality reduction techniques is **Principal Component Analysis (PCA)**. PCA works by transforming the original features into a new set of variables called principal components, which capture the directions of maximum variance in the data. These components are ranked by importance, allowing less informative features to be discarded while retaining the essential structure of the dataset.

A key motivation for using PCA is the **curse of dimensionality**, where increasing the number of features leads to sparse data and decreased model effectiveness. Additionally, real-world datasets often contain noise and redundant information. PCA helps address this through **noise reduction**, improving efficiency and enabling models to generalise better without significant loss of information.

How PCA Works

Principal Component Analysis (PCA) is a dimensionality reduction technique based on the idea that **variance represents important information** in a dataset. Features with higher variance tend to capture meaningful patterns, while those with lower variance often contribute less to distinguishing between observations. PCA aims to identify and retain these important patterns while reducing the number of features.

PCA works by transforming the original variables into a new set of variables called **principal components**. These components are linear combinations of the original features and are arranged in order of importance. The first principal component captures the maximum variance in the data, the second captures the next highest variance while remaining orthogonal (independent) to the first, and so on. This ensures that each component provides new, non-redundant information.

Mathematically, PCA involves computing the **covariance matrix** of the standardised data and extracting its **eigenvectors** and **eigenvalues**. Eigenvectors define the directions of the new axes, while eigenvalues indicate how much variance each component captures. By selecting the top components with the highest eigenvalues, PCA reduces dimensionality while preserving the most significant information in the dataset.

Experimental Focus

This tutorial focuses on analysing how the number of principal components in Principal Component Analysis (PCA) affects both information retention and model performance. While PCA is widely used for dimensionality reduction, selecting the appropriate number of components is a critical decision that directly impacts the effectiveness of the model.

In this experiment, the number of principal components is systematically varied, and two key aspects are evaluated. First, the amount of variance retained is measured to understand how much information from the original dataset is preserved. Second, the performance of a classification model, specifically Logistic Regression, is assessed using accuracy as the evaluation metric.

The objective is to identify the trade-off between dimensionality reduction and predictive performance. Using too few components may result in significant information loss, while using too many may reduce the benefits of simplification. By analysing both variance and accuracy together, this study aims to determine the optimal number of components that balances efficiency and model performance.

Methodology

This study follows a structured approach to evaluate the impact of varying the number of principal components on data representation and classification performance.

First, the **Iris dataset** is selected due to its simplicity, well-balanced classes, and suitability for visualisation. The dataset contains multiple features describing different characteristics of iris flowers, making it appropriate for demonstrating dimensionality reduction.

Next, the data is **standardised** to ensure that all features have a mean of zero and a standard deviation of one. This step is essential because PCA is sensitive to the scale of the input features, and unscaled data can bias the results.

After preprocessing, **Principal Component Analysis (PCA)** is applied to the dataset. The number of components is varied systematically, and for each configuration, the transformed dataset is obtained. The **explained variance ratio** is calculated to determine how much information is retained by the selected components.

To evaluate performance, the dataset is split into training and testing sets, and a **Logistic Regression model** is trained on the PCA-transformed data. The model's accuracy is then measured for each number of components.

Finally, results are visualised using plots to analyse trends in variance retention and classification accuracy, enabling identification of the optimal number of components.

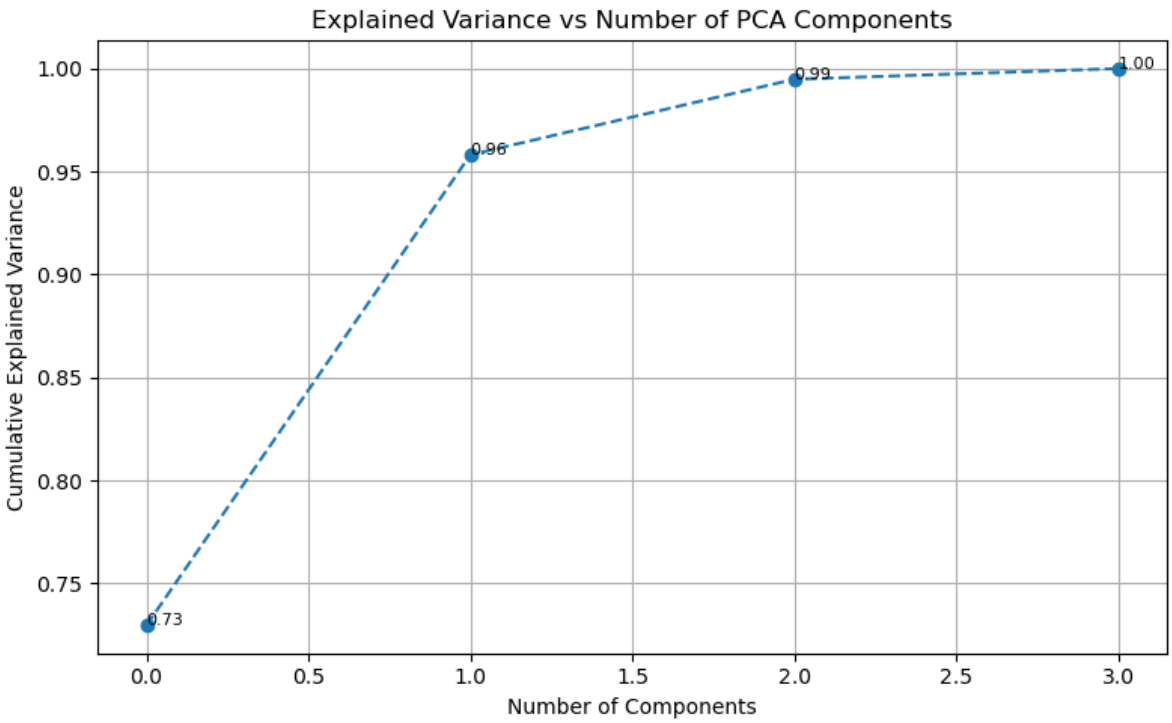
Key Visualisations

To effectively demonstrate the impact of Principal Component Analysis (PCA), three key visualisations are used in this study. These plots help in understanding how dimensionality reduction affects both data representation and model performance.

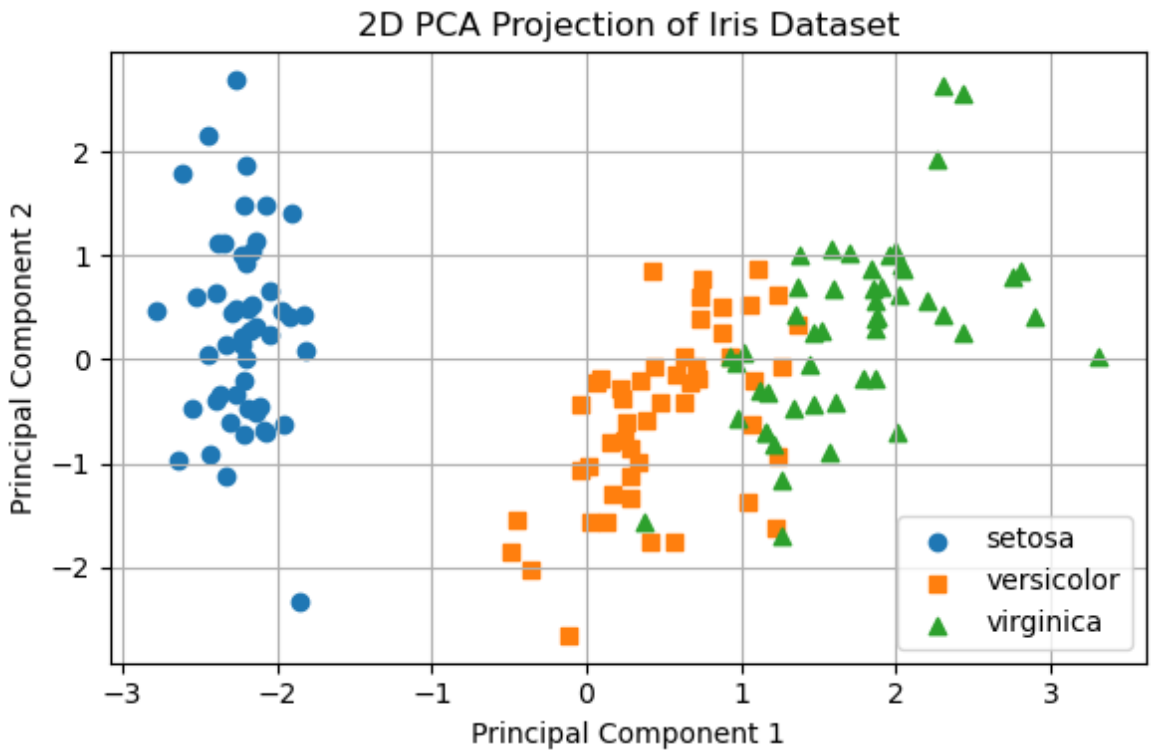
Explained Variance Plot

This plot shows the relationship between the number of principal components and the **cumulative variance retained**. The x-axis represents the number of components, while the y-

axis shows the proportion of total variance captured. This visualisation is important for identifying the **"elbow point"**, where adding more components results in only marginal gains in explained variance. It helps determine the optimal number of components needed to preserve most of the information.



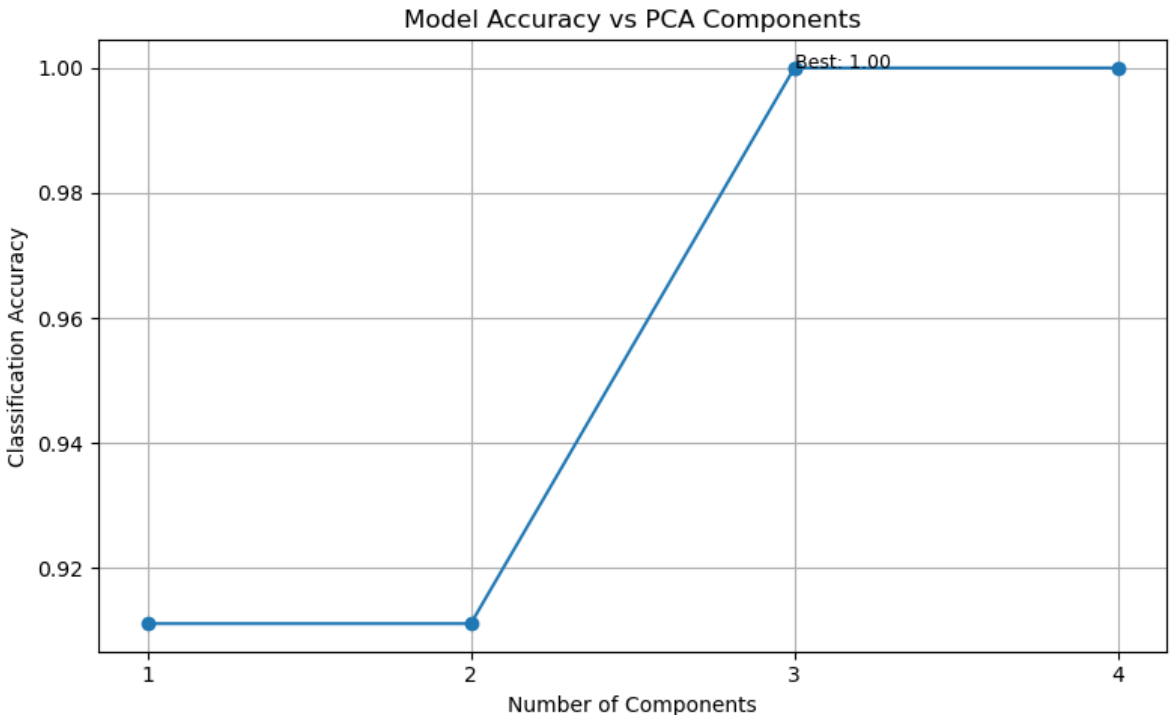
2D PCA Scatter Plot



In this plot, the dataset is projected onto the first two principal components. It provides a visual representation of how well the data points are separated into different classes. This helps in understanding whether PCA preserves meaningful structure and patterns in the data, especially for classification tasks

Accuracy vs Components Graph

This graph illustrates how classification accuracy changes as the number of principal components varies. It highlights the trade-off between reducing dimensionality and maintaining model performance, helping to identify the most effective number of components.



Results and Analysis

The results of this experiment demonstrate a clear relationship between the number of principal components, variance retention, and classification performance. When only a small number of components is used, the model retains limited variance from the original dataset. This leads to a loss of important information, which negatively affects the model's ability to accurately classify data points. As a result, classification accuracy is relatively low in this case.

As the number of principal components increases, the cumulative explained variance also increases, meaning more information is preserved. This generally leads to improved model performance, as the classifier has access to more meaningful features. However, the improvement in accuracy is not linear. After a certain number of components, the increase in explained variance becomes minimal, indicating diminishing returns.

Using too many components reduces the benefits of PCA, as the dataset becomes closer to its original high-dimensional form. This increases computational cost without significantly improving accuracy.

Overall, the results show that there is an optimal number of components where most of the variance is retained while maintaining high classification performance. Selecting this balance is essential for achieving efficient and effective machine learning models.

Conclusion

This study demonstrates that Principal Component Analysis (PCA) is an effective technique for reducing dimensionality while retaining important information. By transforming the original features into principal components, PCA captures the directions of maximum variance, reducing noise and improving computational efficiency. The experiment shows that selecting too few components leads to information loss and lower accuracy, while too many components offer minimal additional benefit. By analysing explained variance and model performance, an optimal number of components can be identified. Overall, PCA enables simpler, more interpretable datasets without significantly compromising predictive accuracy, making it a valuable tool for machine learning.

References

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